MONTE CARLO PRICING OF A EUROPEAN CALL IN A **HYBRID EQUITY-SHORT RATE** MODEL

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The Hybrid model and the Closed-form Benchmark

The Hybrid Model

- Combine a lognormal equity dynamics with a stochastic short interest rate
- Short rate modeled as a Vasicek mean-reverting process
- The stock and short-rate dynamics are driven by correlated Brownian motions with coefficient ρ∈[−1,1]

$$\begin{cases} \dfrac{dS_t}{S_t} = r_t\,dt + \sigma_S\,dW_S^{\mathbb{Q}}(t), \\ dr_t = \kappa(\theta - r_t)\,dt + \sigma_r\,dW_r^{\mathbb{Q}}(t) \end{cases}$$

Equity and Short-Rate Dynamics under Q

- Under the risk-neutral measure Q the arbitrage-free price of the European call option is given by :

$$C_0 = \mathbb{E}^{\mathbb{Q}}\left[\exp\left(-\int_0^T r_t\,dt\right)(S_T - K)^+\right]$$

Closed- form benchmark

- Black–Scholes–type closed-form solution is available for the hybrid model
- p(0,T) is the Vasicek zero-coupon bond price and Σ^2 is an effective variance

$$C_0 = S_0\,N(\tilde{d}_1) - K\,p(0,T)\,N(\tilde{d}_2)$$

$$\tilde{d}_1 = \frac{\ln\left(\frac{S_0}{Kp(0,T)}\right) + \frac{1}{2}\Sigma^2}{\sqrt{\Sigma^2}}, \qquad \tilde{d}_2 = \tilde{d}_1 - \sqrt{\Sigma^2}$$

Given this setup, our goal is to price the European call using Monte Carlo and assess accuracy through comparison with the closed-form solution!

Monte Carlo Methodology

1 Time grid and correlated innovations

The time interval is discretized on a uniform grid, and at each step, two independent standard normal random variables Z1,Z2 N(0,1) are generated and mapped into correlated Brownian increments through :

$$\Delta W_S = \sqrt{\Delta t}\, Z_1, \qquad \Delta W_r = \sqrt{\Delta t}\left(\rho Z_1 + \sqrt{1-\rho^2}\, Z_2\right)$$

2 Discretization of the short rate

The short rate is simulated using an Euler discretization of the Vasicek model, allowing for Gaussian dynamics and possible negative rates

$$\hat{r}_{t_i} = \hat{r}_{t_{i-1}} + \kappa(\theta - \hat{r}_{t_{i-1}})\Delta t + \sigma_r\, \Delta W_r, \qquad i = 1, \ldots, m$$

3 Conditionally exact step for the equity

Conditional on the simulated short rate, the equity price is updated using an exact lognormal step, reducing discretization error relative to a full Euler scheme

$$\hat{S}_{t_i} = \hat{S}_{t_{i-1}} \exp\left[\left(\hat{r}_{t_{i-1}} - \frac{1}{2}\sigma_S^2\right)\Delta t + \sigma_S\, \Delta W_S\right], \qquad i = 1, \ldots, m$$

4 Pathwise discounting and MC estimator

For each simulated path, the payoff is discounted using a Riemann-sum approximation of the short-rate integral, and the option price is obtained as the Monte Carlo estimator

$$\hat{V}^{(j)} = \hat{D}^{(j)}(\hat{S}_T^{(j)} - K)^+, \qquad \hat{D}^{(j)} = \exp\left(-\sum_{i=0}^{m-1} \hat{r}_{t_i}^{(j)} \Delta t\right)$$

$$\hat{C}_0 = \frac{1}{N}\sum_{j=1}^{N} \hat{V}^{(j)}$$

Principal Numerical Issues

1. Two error sources: time discretization error and Monte Carlo sampling error
2. Euler discretization of the short rate introduces the main time-step bias through stochastic discounting
3. Equity simulated with a conditionally exact lognormal step, reducing discretization error
4. The option price depends on the joint distribution of equity and stochastic discount factor, making correlation relevant
5. The Vasicek model allows negative interest rates, a structural feature of Gaussian short-rate models

Implementation and programming considerations

```
r = np.zeros((N_sim, m+1))
S = np.zeros((N_sim, m+1))
```

Vectorized Monte Carlo implementation to reduce computational time

```
Z1 = np.random.normal(size=N_sim)
Z2 = np.random.normal(size=N_sim)
dW_S=sqrt_dt*Z1
dW_r=sqrt_dt*(rho*Z1+np.sqrt(1-rho**2)*Z2)
```

Explicit construction of correlated Brownian motions

```
payoff = np.maximum(S[:, -1] - K, 0.0)

int_r_dt = np.sum(r[:, :-1] * dt,axis=1)
disc = np.exp(-int_r_dt)

pv = disc * payoff
pv_call = pv.mean() #MC estimator
```

Pathwise discounting and Monte Carlo estimator

Hybrid discretization: Euler for the short rate and exact lognormal step for equity

```
r[:, i+1] = r[:, i] + k*(theta - r[:, i])*dt + sigmar*dW_r
S[:, i+1] = S[:, i] * np.exp((r[:, i] - 0.5*sigmas**2)*dt + sigmas*dW_S)
```

Numerical results and validation

Model Setup

```
m = 250          # Yearly trading days
T = 1            # Horizon: One year
dt = T/m         # discretization delta
N_sim= 100000
r0 = 0.0364      # American T-Bill 27/12
theta = 0.04     # Long-Term Interest rate
k = 0.2          # Mean Reversion coefficient.

sigmar = 0.02    # volatility of interest rates
sigmas = 0.25    # volatility of underlying S
S0 = 140         # Starting Price of the underlying
K = 100          # strike price
rho = 0.25        #Correlation coefficient
```

Model Results

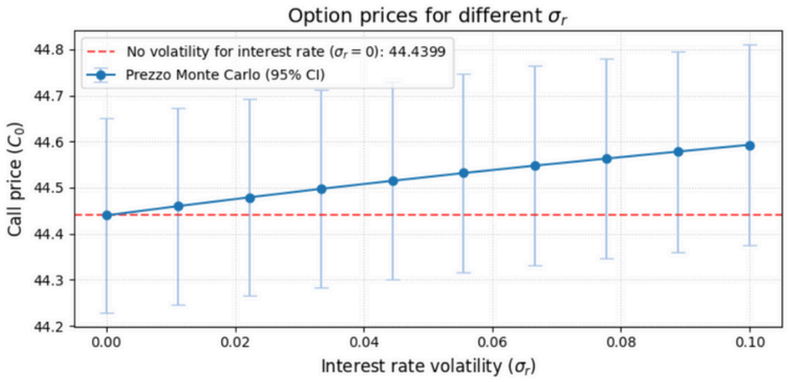
Monte Carlo price = 44.475234
95% confidence interval: [44.2622, 44.6883]
Closed-form benchmark: = 44.501294

Model Validation

- The analytical value lies within the Monte Carlo confidence interval
- Discretization bias at Δt=1/250 is small relative to sampling error

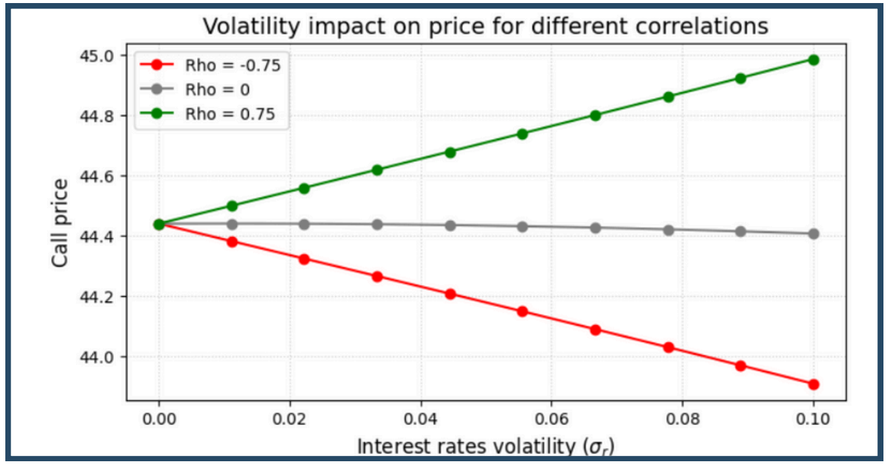
Sensitivity Analysis

Sensitivity to Interest-Rate Volatility

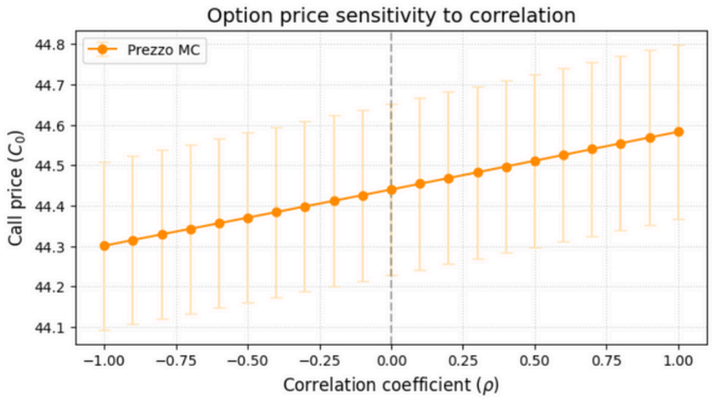


With all other parameters fixed and ρ=0.25, the option price increases mildly with σ_r, showing narrow confidence intervals and reflecting higher variability of the integrated short rate

Joint Effect of Interest-Rate Volatility and Correlation



Sensitivity to Correlation



Keeping all other parameters fixed, the option price increases approximately linearly with the equity–interest rate correlation ρ, reflecting the direct impact of correlation on the distribution of discounted payoffs

The **joint effect** of interest-rate volatility and correlation is analyzed by varying σ_r across representative values of ρ. The results show that **the sign of ρ can reverse the impact of the volatility**: prices increase with σ_r for positive correlation, decrease for negative correlation, and remain nearly flat when ρ is near to 0

An empirical illustration : European Call on Apple (AAPL)

```
--- Market Data AAPL ---
Underlying price (S0): 267.26
Historical volatility (sigma_s): 32.19%
------------------------------
 Call Price (Monte Carlo): 37.656164
Radius of Monte Carlo Estimator:  0.3963
MC estimator 95% confidence interval: [ 37.2599 , 38.0524 ]
 Call Price (Closed form): 37.930857
Absolute error: 0.274693
```

Computed on 6th January 2026 with closed price until 5th January

This empirical exercise shows that, even with market-based equity inputs and simplified rate parameters, the Monte Carlo estimator remains consistent with the analytical benchmark!

Conclusion

- **Monte Carlo prices match the closed-form benchmark within confidence intervals**
- **Interest-rate volatility and correlation materially affect option values**
- **The framework is numerically stable and practically applicable**